

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                        |
|----------------------|------------------------|
| United States Patent | 12405785               |
| Kind Code            | B1                     |
| Date of Patent       | September 02, 2025     |
| Inventor(s)          | Pathak; Avanish et al. |

---

### Phased deployment

---

#### Abstract

Techniques are described for deploying software applications associated with document management. The techniques may include a system comprising processing circuitry implemented by one or more processors, the processing circuitry having access to a memory. The processing circuitry may be configured to determine, for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications. The processing circuitry may further be configured to output instructions to execute the current version of the widget in response to receiving a first request for the software application. The processing circuitry may further be configured to output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget.

---

**Inventors:** Pathak; Avanish (Rochester, NY), Peterson; Mark (Bellevue, WA), Krishna; Bhargav (Snohomish, WA), Singh; Vikas (San Jose, CA)

**Applicant:** DocuSign, Inc. (San Francisco, CA)

**Family ID:** 96882077

**Assignee:** Docusign, Inc. (San Francisco, CA)

**Appl. No.:** 18/362762

**Filed:** July 31, 2023

---

#### Publication Classification

**Int. Cl.:** G06F8/71 (20180101); G06F8/65 (20180101)

**U.S. Cl.:**

**CPC** G06F8/71 (20130101); G06F8/65 (20130101);

## Field of Classification Search

USPC: None

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC          |
|--------------|-------------|---------------|----------|--------------|
| 8676943      | 12/2013     | Parthasarathy | 709/221  | G06F 9/5061  |
| 9158577      | 12/2014     | Lipscomb      | N/A      | G06F 8/656   |
| 10019255     | 12/2017     | Greenfield    | N/A      | G06F 11/1433 |
| 11269616     | 12/2021     | Schumaker     | N/A      | G06F 8/65    |
| 2014/0013315 | 12/2013     | Genevski      | 717/170  | G06F 8/60    |
| 2014/0379901 | 12/2013     | Tseitlin      | 709/224  | H04L 41/5025 |
| 2021/0234887 | 12/2020     | Landman       | N/A      | H04L 63/1433 |
| 2022/0156053 | 12/2021     | Shaastry      | N/A      | G06F 8/433   |

### FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC        |
|---------------|------------------|---------|------------|
| 114625410     | 12/2021          | CN      | N/A        |
| WO-2023143237 | 12/2022          | WO      | G06F 21/12 |

### OTHER PUBLICATIONS

U.S. Appl. No. 18/345,799, filed Jun. 30, 2023, naming inventors Pathak et al. cited by applicant  
U.S. Appl. No. 18/345,815, filed Jun. 30, 2023, naming inventors Pathak et al. cited by applicant  
Zakas, "Scalable JavaScript Application Architecture", Sep. 7, 2009, 108 pp., URL: <https://ffconf.org/talks/scalable-javascript-application-architecture/>. cited by applicant

---

*Primary Examiner:* Wang; Philip

*Attorney, Agent or Firm:* Shumaker & Sieffert, P.A.

---

## Background/Summary

### TECHNICAL FIELD

(1) This disclosure relates generally to management of a software application.

### BACKGROUND

(2) Continuous integration and continuous deployment (CI/CD) environments implemented by software development teams typically include utilities for software developers to upload and deploy a software application. Current CI/CD environments generally involve software applications developed by various teams. CI/CD environments generally require a server hosting the software application to restart every time a new version of the software application is deployed.

## SUMMARY

(3) Aspects of the present disclosure describe techniques for deploying a software application associated with document management. Software applications associated with document management may include user experiences related to managing electronic or digital documents (e.g., generate a document, sign a document, etc.). Software developers may create, update, integrate, deploy, or otherwise develop software applications via a continuous integration and continuous deployment (“CI/CD”) environment. Software applications are deployed when the CI/CD environment allows client devices to request and access files and data included in the software applications. Using the CI/CD environments may help to implement a common knowledge base that may help to ensure compatibility of all software components (e.g., widgets and/or plugins) used in software application development and runtime deployment. In this way, the techniques allow new versions of code developed by software developers to be seamlessly deployed with previously deployed code at runtime.

(4) The techniques described herein help to provide software developers flexibility when deploying new versions of a component of a software application (e.g., a new version of a widget). For example, a single server or a computing system may implement a phased deployment, where the system responds to each request for a software application such that the responses to the requests specify a target ratio of a new version and currently deployed version of a widget. The system may, for example, progressively increase the target ratio based on a time window defined by software developers of a new version of a widget. The system may progressively increase the target ratio until all client devices receive instructions to execute a new version of the widget at the close of the time window. In this way, the system may efficiently manage versions of widgets executing on client devices without additional computational and/or network resources such as load balancers. The system may monitor execution of a new version of a widget throughout the time window to take a particular action (e.g., roll-back deployment of the new version of the widget, increase target ratio at a faster rate, etc.) or make a suggestion (e.g., issue an alert to a software developer device indicating one or more performance criteria have not been met) based on a relationship of monitored metrics and configurable criteria.

(5) In one example, the present disclosure describes a system for deploying a software application associated with document management. The system may comprise processing circuitry implemented by one or more processors, the processing circuitry having access to a memory. The processing circuitry may be configured to determine, for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget. The processing circuitry may further be configured to output instructions to execute the current version of the widget in response to receiving a first request for the software application. The processing circuitry may further be configured to output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request in response to receiving a second request for the software application after the first request.

(6) In another example, the present disclosure describes a method for deploying a software application associated with document management. The method may include determining, by a computing system and for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget. The method may further include in response to receiving a first request for the software application, outputting, by the computing system, instructions to

execute the current version of the widget. The method may further include in response to receiving a second request for the software application after the first request, outputting, by the computing system, instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request.

(7) In another example, the present disclosure describes a computer-readable storage medium comprising instructions that, when executed, configure processing circuitry of a computing system to determine, for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget. The processing circuitry may further be configured to output instructions to execute the current version of the widget in response to receiving a first request for the software application. The processing circuitry may further be configured to output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request in response to receiving a second request for the software application after the first request.

(8) The details of one or more examples of the techniques of this disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques will be apparent from the description and drawings, and from the claims.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagrams illustrating an example computing environment for managing deployment of software applications, in accordance with techniques of this disclosure.

(2) FIG. 2 is a block diagram illustrating an example computing environment with example details of a computing system managing deployment of software applications, in accordance with the techniques of this disclosure.

(3) FIG. 3 is a block diagram illustrating an example computing system for managing software applications, in accordance with the techniques of this disclosure.

(4) FIG. 4 is a conceptual diagram illustrating an example deployment lifecycle of a new version of a widget for a software application, in accordance with techniques of this disclosure.

(5) FIG. 5 is a conceptual diagram illustrating an example configuration file, in accordance with the techniques of this disclosure.

(6) FIG. 6 is a block diagram illustrating an example consumer computing device, in accordance with the techniques of this disclosure.

(7) FIG. 7 is a conceptual diagram illustrating example contextual information used to load a software application, in accordance with the techniques of this disclosure.

(8) FIG. 8 is a block diagram illustrating an example developer computing device, in accordance with the techniques of this disclosure.

(9) FIG. 9 is a conceptual diagram illustrating an example deployment configuration for a new version of a widget, in accordance with techniques of this disclosure.

(10) FIG. 10 is a flow chart illustrating an example mode of operation for managing deployment of software applications, in accordance with the techniques of this disclosure.

(11) FIG. 11 is a flow chart illustrating an example mode of operation for deploying a new version of a widget, in accordance with the techniques of this disclosure.

(12) FIG. 12 is a flow chart illustrating an example mode of operation for managing the deployment of a new version of a widget, in accordance with the techniques of this disclosure.

(13) Like reference characters denote like elements throughout the text and figures.

## DETAILED DESCRIPTION

(14) In general, aspects of the present disclosure describe techniques for deploying a software application associated with document management. Software applications associated with document management may include user experiences related to managing electronic or digital documents (e.g., generate an electronic document, sign an electronic document, etc.). Software applications may be deployed with multiple widgets. A widget may be a collection of software code used by the software application to provide the user experience (e.g., create a new Portable Document Format (PDF) file, open a PDF file, etc.). For example, a widget may include a snippet of JavaScript® logic that runs in a web browser. Software developers may create, update, integrate, deploy, or otherwise develop widgets for software applications via a continuous integration and continuous deployment (“CI/CD”) environment.

(15) Software developers may use the CI/CD environment to deploy a new version of a software application or a software application component (e.g., widget) by instructing consumer devices to load code files associated with the new versions of the software application or software application component. Some CI/CD environments may redirect consumer-related software application requests (e.g., network-based requests made for the software application) by rebooting a server hosting the software application to include the new version of the software application. Some CI/CD environments may reduce or eliminate rebooting a server of a CI/CD computing system (e.g., a cloud-based server and/or content delivery network) by, for example, implementing a load balancer to redirect software application requests to another server hosting the new version of the software application. For example, a CI/CD computing system may include a load balancer developed to redirect consumer traffic from a server of the CI/CD computing system hosting a current version of a widget to a different server of the CI/CD computing system hosting a new version of a widget.

(16) The techniques described herein may include a CI/CD computing system that may dynamically deploy new versions of a software application component (e.g., widget) during runtime or execution of software applications implementing the software application component. The CI/CD computing system may dynamically deploy new versions of widgets without restarting a server hosting the software application component and without using a load balancer. In some instances, the CI/CD computing system may specify, based on a target ratio and a time window, either the new version of the widget or the current version of the widget in responses to requests for software applications implementing the widget. The CI/CD computing system may dynamically deploy a new version of a widget by progressively sending instructions to requesting client devices to execute the new version of the widget. For example, the CI/CD computing system may send instructions to execute a new version of a widget by specifying an identifier or address (e.g., uniform resource indicator) associated with the new version of the widget in a configuration file included in the instructions to execute a requested software application implementing the widget. The CI/CD computing system may send instructions to execute the new version of the widget based on a target ratio associated with a particular phase of the time window and in response to metrics associated with execution of the new version of the widget satisfying criteria. In this way, the CI/CD computing system may manage deployment of new versions of software application components without implementing load balancers. The techniques described herein may avoid a CI/CD computing system statically implementing load balancers for deployment of new versions of widgets during build time (e.g., steps associated with preparing software application code for demonstration or production). The CI/CD computing system supports implementation of a singular server to support any number of versions instead of relying on a unique server instance for each version.

(17) The techniques described herein may provide one or more technical advantages that realize one or more practical applications. For example, a computing system may operate as a common

knowledge base to dynamically deploy new versions of a software application or software application component (e.g., widget). The computing system may dynamically deploy new versions of software application components by progressively sending configuration information associated with the new versions of the software application components during a time window. In some instances, the computing system may determine a total number of requests for the widget during a particular time within a time window. The computing system may determine a number of requests selected as recipients of a new version of the widget based on a target ratio associated with the particular time within the time window. For example, the computing system may determine a total number of requests for a widget is 100 during a middle phase of the time window. The computing system may configure the middle phase of the time window to define a target ratio of 0.5. The computing system may determine that during the middle phase of the time window, 50 requests for the widget receive instructions to execute the current version of the widget and the other 50 requests for the widget receive instructions to execute the new version of the widget. In some examples, the computing system may implement a machine learning model to determine the target ratio based on obtained metrics associated with execution performance of the new version of the widget.

(18) The computing system may receive an input from software developers specifying a time window that defines a period in time consumer devices should execute a new version of a software application component. In some instances, the computing system may receive an input of a time window specifying a start date and/or time and an end date and/or time. In some examples, the computing system may partition the time window into phases and assign a target ratio to each phase. For example, the computing system may receive an input specifying that a new version of a widget will begin deployment at 5:00 pm and end at 6:00 pm on the same day. The computing system may partition the one-hour time window into four equally spaced phases and assign a target ratio of responses including the new version of the application divided by a total number of requests for the widget. The computing system, for example, may assign a target ratio of zero (no instructions to execute the new version of the widget) before the time specified in the time window begins. The computing system may assign a target ratio of 0.25 to the first phase (25% of requests for the widget receive instructions to execute the new version of the widget), 0.5 for the second phase (50% of requests for the widget receive instructions to execute the new version of the widget), 0.75 for the third phase (75% of requests for the widget receive instructions to execute the new version of the widget), and 1 for the final phase (100% of requests for the widget receive instructions to execute the new version of the widget).

(19) The computing system may respond to consumer requests for a software application component with configuration information associated with the new version of the software application component and instructions to execute the new version of the software application component. The consumer device requesting the software application component may not experience interruptions when requesting the software application component when the new version of the software application component is deployed because the computing system sends instructions to execute the new version of the software application based on configuration information associated with the new version of the software application. The computing system may reduce a processing burden associated with a server or software repository hosting software application components. For example, a software repository hosting widgets is not necessarily required to track versions of widgets because the computing system may dynamically manage versions of widgets with one or more configuration files. The software repository may not necessarily be required to restart when deploying the new version of the software application component when the software repository does not use computational resources in the management and deployment of widgets. The computing system may further reduce computational resources by managing progressive deployment of new versions of widgets without any load balancer to redirect consumer traffic to a new version of the widget. The computing system may be implemented as a

single server that can individually manage deployment of new versions of widgets without implementing a load balancer to redirect traffic to an additional server. The computing system may efficiently manage deployment of a new version of a widget with configuration information and instructions sent in a response to a request for a software application that implements the widget. The computing system may reduce processing resources associated with managing software application deployment by, for example, redirecting consumer traffic to a new version of the software application component during runtime (e.g., while a software application is being executed by a consumer device).

(20) The computing system may help to provide software developers flexibility in managing deployment of a new version of a software application component without having to implement any load balancing techniques during build time. Software developers may build a new version of a widget, locally test the new version of the widget, integrate the new version of the widget in software applications, and stage the new version of the widget by executing quality assurance and/or security tests on the new version of the widget. The computing system may provide software developers an option to phase deployment of a new version of a widget during a demonstration or production stage associated with the lifecycle of the new version of the widget. The computing system may enable software developers to deploy the new version of the widget to a subset of client devices. The computing system may monitor runtime performance metrics associated with the new version of the widget executing on the subset of client devices. In this way, the computing system may minimize risk of unknown or undetected errors caused by the new version of the widget because the computing system has only instructed a subset of client devices to execute the new version of the application. The computing system may determine whether the new version of the widget results in an enhancement over a current or previous version of the widget based on the obtained metrics. In this way, the computing system provides software developers the ability to monitor execution of a new version of a widget only on a subset of client devices to make determinations on whether to roll-back or speed up deployment of the new version of the widget.

(21) FIG. 1 is block diagram illustrating an example computing environment **10** for managing deployment of software applications, in accordance with techniques of the disclosure. In the example of FIG. 1, computing environment **10** includes computing system **100**, developer computing device **150**, consumer computing device **160**, and software repository **170**. Computing environment **10** may represent a CI/CD environment for developers to write and release software applications. Computing environment **10** may include a starter kit, utilities, standard development kits, application programming interfaces, or other developer resources. Computing system **100** may manage the developer resources to help developers upload code to a repository that is compatible with other code in the repository. In some examples, computing system **100** may represent a cloud computing system that provides one or more services via network **101**. Computing system **100** may include a collection of hardware devices, software components, and/or data stores that can be used to implement one or more applications or services provided to developer computing device **150** and consumer computing device **160** via network **101**. Computing system **100** may represent a cloud-based implementation. As used herein, client device may refer to developer computing device **150**, consumer computing device **160**, or a device that performs the function of both developer computing device **150** and consumer computing device **160**.

(22) Network **101** may include the Internet or may include or represent any public or private communications network or other network. For instance, networks **101** may be a cellular, Wi-Fi®, ZigBee®, Bluetooth®, Near-Field Communication (NFC), satellite, enterprise, service provider, and/or other type of network enabling transfer of data between computing systems, servers, computing devices, and/or storage devices. One or more of such devices may transmit and receive data, commands, control signals, and/or other information across network **101** using any suitable communication techniques. Network **101** may include one or more network hubs, network switches, network routers, satellite dishes, or any other network equipment. Such network devices

or components may be operatively inter-coupled, thereby providing for the exchange of information between computers, devices, or other components (e.g., between one or more client devices or systems and one or more computer/server/storage devices or systems). Each of the devices or systems illustrated in FIG. 1 may be operatively coupled to network 101 using one or more network links. The links coupling such devices or systems to network 101 may be Ethernet, Asynchronous Transfer Mode (ATM) or other types of network connections, and such connections may be wireless and/or wired connections. One or more of the devices or systems illustrated in FIG. 1 or otherwise on network 101 may be in a remote location relative to one or more other illustrated devices or systems.

(23) Computing system 100, developer computing device 150, consumer computing device 160, and software repository 170 may send and receive data via network 101. Data exchanged over the network 101 may be represented using any suitable format, such as hypertext markup language (HTML), extensible markup language (XML), or JavaScript Object Notation (JSON). In some aspects, network 101 may include encryption capabilities to ensure the security of documents. For example, encryption technologies may include secure sockets layers (SSL), transport layer security (TLS), virtual private networks (VPNs), and Internet Protocol security (IPsec), among others.

(24) Examples of developer computing device 150 and consumer computing device 160 may include, but are not limited to, portable, mobile, or other devices, such as mobile phones (including smartphones), wearable computing devices (e.g., smart watches, smart glasses, etc.) laptop computers, desktop computers, tablet computers, smart television platforms, server computers, mainframes, infotainment systems (e.g., vehicle head units), etc. Developer computing device 150 and consumer computing device 160 may be examples of client devices that may interact with computing system 100 to upload, use, or otherwise access software applications managed by computing system 100.

(25) Developer computing device 150 may include command line interface 152. Developer computing device 150 may be operated by a software developer who may upload and access code for a software application based on configuration information received from computing system 100. Command line interface 152 may provide an interactive environment for developers to develop and deploy software applications. Command line interface 152 may include, but is not limited to, a shell terminal or application programming interface managed by computing system 100. As used herein, a shell terminal may refer to a software program that acts as command-line interpreter for services related to an application provided by computing system 100. For example, command line interface 152 may process commands (e.g., user inputs) received by developer computing device 150 to generate instructions and output, via network 101, the instructions to computing system 100 and/or to software repository 170. For instance, developer computing device 150 may use command line interface 152 to upload software code to software repository 170.

(26) Developer computing device 150 may pull software applications and resources associated with the software applications from software repository 170. Software repository 170 may include, but is not limited to, a content delivery network, cloud storage systems, or any collection of databases used to store and send software files. For example, software repository 170 may be a content delivery network that implements Microsoft Solutions Framework®.

(27) Consumer computing device 160 may include shell interface 162. Consumer computing device 160 may be a computing device operated by a consumer of software applications managed by computing system 100. Shell interface 162 may include, but is not limited to, a web browser or user interface that may request data from computing system 100 and software repository 170 via network 101. Consumer computing device 160 may request access to a software application with shell interface 162 (e.g., a browser shell interface or application interface). For example, consumer computing device 160 may request the software application to manage electronic documents with shell interface 162. In some systems, CI/CD environments may require a server hosting the software application to restart when deploying a new version of the software application. This may



result in interruptions of a user's experience using the software application. The techniques described herein may apply a common knowledge base (e.g., computing system **100**) to regulate network traffic to new versions of software applications. In this way, the techniques help to reduce or eliminate restarting a host of a software application when a new version of the software application is deployed. The techniques described herein may help to improve an availability of software repository **170**, which may potentially improve a user's experience using the software applications. The techniques described herein may send consumer computing device **160** instructions to execute a new version of a software application without interrupting the experience of a user operating consumer computing device **160** while executing the new version of the software application.

(28) In the example of FIG. **1**, computing system **100** may include one or more servers locally managed by a software administrator or a distributed or cloud-based computing system managed by the software administrator. Computing system **100** may include one or more computing devices, servers, databases, etc. communicably coupled to perform the techniques described herein. Computing system **100** may include management controller **102** and storage devices **104**. Management controller **102** may be a module with instructions to manage new and updated software applications. For example, management controller **102** may be configured to upload and/or update configuration files associated with widget configuration information used to implement a software application. Management controller **102** may store new and/or updated configuration files associated with widget configuration information in storage devices **104**. Storage devices **104** may represent one or more physical or virtual computer and/or storage devices that include or otherwise have access to storage media. Such storage media may include one or more of Flash drives, solid state drives (SSDs), hard disk drives (HDDs), forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories, and/or other types of storage media used to support computing system **100**. Functionality of management controller **102** and storage devices **104**, as described herein, may be distributed across multiple computing devices within the distributed or cloud-based environment of computing system **100**.

(29) In accordance with the techniques of the disclosure, computing system **100** may dynamically redirect consumer requests for a software application to include a new version of a widget. Management controller **102** of computing system **100** may generate a response to a request from consumer computing device **160** for a software application. Management controller **102** may generate a response that includes instructions to execute the software application based on configuration information indicating a new version of a widget used in the execution of the software application. In some instances, management controller **102** may progressively include configuration information associated with the new version of the widget in the instructions to execute the software application based on a configurable time window. For example, consumer computing device **160** may execute a software application with a previous version of the widget until computing system **100** selects consumer computing device **160** as a recipient of configuration information associated with the new version of the widget. Computing system **100** may apply a machine learning model, or make inferences, to identify runtime characteristics of software applications executing with the new version of the widget. In some instances, computing system **100** may automatically roll back deployment of the new version of the widget or alert a team of software developers in response to the identified runtime characteristics not satisfying performance, security, or other quality assurance requirements. In this way, computing system **100** may monitor runtime characteristics of the new version of the widget executing on other consumer computing devices prior to instructing consumer computing device **160** to execute the new version of the widget.

(30) In operation, computing system **100** may monitor software repository **170** for new versions of widgets used in the execution of software applications. Developer computing device **150** may use

command line interface **152** to upload a new version of a widget to software repository **170**. For instance, a user may provide, using command line interface **152**, code for a new widget, perform testing and validation of the code, and, upon confirming that the code has satisfied the testing and validation, upload the code for storage by software repository **170** with an indication of the widget and a version for the code. In some instances, command line interface **152** may upload a build artifact of the new version of the widget to software repository **170**. Command line interface **152** may upload the new version of the widget in a standardized format as a build artifact to ensure the new version of the widget is compatible with all other widgets managed by computing system **100**.

(31) In some instances, computing system **100** may generate a configuration file for the new version of the widget that includes configuration information associated with the new version of the widget. In some examples, computing system **100** may generate a deployment configuration file corresponding to an environment associated with the current integration and deployment lifecycle of the new version of the widget. For example, computing system **100** may generate a deployment configuration file for a production environment. A production environment may represent a stage in a widget lifecycle associated with the widget being executed on consumer computing device **160**. Computing system **100** may only generate a deployment configuration file for the new version of the widget to be implemented in the production environment in response to the new version of the widget satisfying quality assurance and/or security requirements. In some instances, computing system **100** may include a preliminary threshold associated with the compatibility of the new version of the widget with the ecosystem supported by computing system **100**.

(32) In some instances, computing system **100** may determine (e.g., by periodically polling or based on an event) a current version of a widget and a next version of the widget to implement a software application. Computing system **100** may determine the current version of the widget and the next version of the widget for a time window. Computing system **100** may determine the current version of the widget and the next version of the widget for a time window that has been defined by developer computing device **150** when uploading the next version of the widget to software repository **170**.

(33) Software repository **170** may be configured to provide either the current version of the widget or the next version of the widget according to application programming interface (API) calls made based on either configuration files associated with the current version of the widget or configuration files associated with the next version of the widget. Computing system **100** may instruct a computing device requesting a software application that implements the widget (e.g., consumer computing device **160**) to make an API call for the current version of the widget based on configuration information associated with the current version of the widget. Computing system **100** may instruct consumer computing device **160** to make an API call for the next version of the widget based on configuration information associated with the next version of the widget. Computing system **100** may instruct consumer computing device **160** to make an API call for the next version of the widget based on a number of previous requests for the widget made at a particular point in the time window defined by developer computing device **150**. For instance, computing system **100** may instruct consumer computing device **160** to make a call for the next version based on a target ratio. Computing system **100** may progressively increase a target ratio throughout the time window. Computing system **100** may determine a target ratio representing a maximum percentage of requests for the widget that will receive instructions to execute the new version of the widget. For example, computing system **100** may have a target ratio of 0.0 at the beginning of the time window, such that no requests for the widget receive instructions to execute the new version of the widget. In some examples, computing system **100** may increase the target ratio proportional to the length of time defined in the time window. For example, computing system **100** may increase the target ratio by 0.01 every minute for a time window defining a 100-minute length of time.

(34) In some instances, computing system **100** may instruct consumer computing device **160** to

make a call for a next version of a widget. Computing system **100** may determine a target ratio for a particular amount of time in the time window. Computing system **100** may determine a target ratio that may include a maximum number of intended recipients of the new version of the widget divided by the total number of requests for the widget. For example, computing system **100** may determine a target ratio of 0.5 at a middle point in the time window. In this example, computing system **100** may determine that the total number of requests for the widget is 101 requests for the widget. Computing system **100** may select 50 out of the 101 requests for the widget as recipients of the new version of the widget as to not exceed the target ratio of 0.5. Conversely, computing system **100** may determine a target ratio with respect to a minimum number of recipients of a current version of the widget. In the example of computing system **100** determining there are 101 requests for the widget, computing system **100** may apply a target ratio of 0.5 representing a minimum number of requests that may receive instructions to call and execute the current version of the widget. Computing system **100** may select 51 out of the 101 requests for the widget a recipient of the current version of the widget as to not go below the target ratio of 0.5. In this way, computing system **100** may actively monitor performance of the execution of the new version of the widget to avoid potential issues affecting a majority of requesting computing devices.

(35) Management controller **102** of computing system **100** may receive a request from consumer computing device **160** to access a software application that may use the new version of the widget. Shell interface **162** may output the request for the software application via network **101**. Management controller **102** may receive the request and generate a response. Management controller **102** may generate a response that includes instructions to execute the software application based on configuration information associated with widgets used in the implementation of the software application. For example, management controller **102** may receive a request to provide a plugin from consumer computing device **160**. In this example, management controller **102** may determine one or more of widgets that provide the plugin. For instance, management controller **102** may determine a plugin and one or more widgets that the configuration information specifies as providing functionality (e.g., a PDF reader, authentication, etc.) for the plugin. In this example, management controller **102** may generate the response to include instructions to execute each of the one or more widgets identified in the configuration information as providing functionality for the plugin. Management controller **102** may send the response to shell interface **162** via network **101**. Shell interface **162** may receive the response and request software code files from software repository **170** based on the instructions included in the response.

(36) Computing system **100** may output instructions to execute the current version of a widget of a software application in response to shell interface **162** requesting the software application. Computing system **100** may output the instructions to execute the current version of the widget based a number of requests for software applications that include the current widget. Computing system **100** may output instructions to execute the current version of the widget based on the time window defined when developer computing device **150** uploaded the new version of the widget to software repository **170**. For example, developer computing device **150** may upload a new version of a widget and define a time window of one hour to start immediately. At the time computing system **100** generates a deployment configuration file with configuration information associated with the new version of the widget, computing system **100** may send instructions to consumer computing device **160** to execute the current version of the widget. In some examples, computing system **100** may divide the time window into discrete phases, such as, for example, four phases of fifteen-minutes for a one-hour time window. Computing system **100** may assign a target ratio to each phase representing the number of recipients to be selected to receive the new version of the widget, divided by the total number of requests for the widget. Computing system **100** may refresh the total number of requests for the widget continuously (e.g., every three seconds). During the first phase (e.g., the first fifteen minutes of the time window), computing system **100** may assign a target ratio of 0.25. Computing system **100** may select 25% of the total requests for the widget as

recipients of instructions to execute the new version of the widget. For example, computing system **100** may receive one-hundred requests for the widget. During the first phase, computing system **100** may respond to 25 (out of the 100) requests for the widget with instructions and configuration information associated with the new version of the widget and send a response to the other 75 requests with instructions to load a current or previous version of the widget. During the second phase (e.g., the second interval of fifteen minutes of the time window), computing system **100** may determine the number of requests for software applications that include the widget and allocate 50% of those requests as recipients of instructions to execute the new version of the widget. During the third phase (e.g., the third interval of fifteen minutes of the time window), computing system **100** may determine the number of requests for software applications that include the widget and allocate 75% of those requests as recipients of instructions to execute the new version of the widget. During the final phase (e.g., the last fifteen minutes of the time window), computing system **100** may determine the number of requests for software applications that include the widget and allocate 100% of those requests as recipients of instructions to execute the new version of the widget.

(37) In some examples, computing system **100** may divide the time window into phases on the order of seconds. For example, computing system **100** may divide a time window into phases representing one-second. In this example, computing system **100** may increase the number of requests as recipients of instructions to execute the new version of the widget every second. Computing system **100** may increase the number of requests as recipients of instructions to execute the new version of the widget according to a target ratio. Computing system **100** may progressively increase the target ratio in each phase computing system **100** divided the time window into. For example, computing system **100** may divide a time window of 100 seconds into 100 phases representing one-second each. Computing system **100** may increase the target ratio by 0.01 in each phase of the time window. Computing system **100** may select the recipients of instructions to execute the new version of the widget by determining a total number of requests in each phase. In response to computing system **100** determining the total number of requests in a particular phase, computing system **100** may select a number of the requests as recipients of instructions to execute the new version of the widget in such a way as to not exceed the target ratio associated with the particular phase. For example, during the 75.sup.th phase out of 100 phases with a target ratio of 0.75, computing system may determine a total number of requests for the widget is 14. Computing system **100** may select ten out of the 14 requests as recipients of instructions to execute the new version of the widget. In this way, computing system **100** may minimize any potential issues associated with the execution of the new version of the widget on computing devices.

(38) In some instances, computing system **100** may select consumer computing device **160** as a recipient of configuration information associated with a new version of a widget. For example, consumer computing device **160** may be selected as a recipient of instructions to execute the new version of a widget during the final phase of a time window defined when the new version of the widget was uploaded to software repository **170**. In some instances, consumer computing device **160** may not receive instructions to execute the new version of the widget if computing system **100** triggers an automatic roll back of the new version of the widget. For example, computing system **100** may receive metrics from other consumer computing devices executing the new version of the widget during the demonstration and/or production stage of the new version of the widget's lifecycle. Although developer computing device **150** may conduct pre-deployment validation prior to the demonstration or production stage of the new version of the widget, computing system **100** may monitor execution of the new version of the widget on consumer computing device **160** to determine whether quality assurance and/or security requirements are far below a threshold value. In some examples, computing system **100** may determine performance, quality assurance, security, etc. requirements of a new version are not met and take an action to either roll-back or alert a developer of developer computing device **150** that an issue is detected in the deployment of the

new version of the widget.

(39) Computing system **100** may obtain metrics associated with execution of the new version of the widget on consumer computing device **160**. Examples of metrics may include, for example, a number of gate failures, uncaught bugs, API issues, critical security issues, or a particular combination of widget performing poorly. In response to determining that metrics received from computing devices executing the new version of the widget do not satisfy one or more criteria associated with quality assurance and/or security standards, computing system **100** may instruct all computing devices to execute the current version of the widget and stop instructing more computing devices to execute the new version of the widget (e.g., roll-back deployment of the new version of the widget). For example, computing system **100** may roll-back deployment of a new version of a widget as a result of gate failures or software bugs that may be unpredictable and was not encountered during internal testing (e.g., pre-deployment testing, quality assurance testing, security testing, SOC 2 compliance, etc.). In some examples, metrics may include a central processing unit (CPU) performance, memory performance, a graphical processor unit (GPU) performance, battery performance, application launch time, widget load time, number of errors or warnings during runtime execution of the new version of the widget, etc. For example, computing system **100** may receive metrics indicating that a CPU performance (e.g., clock cycles, threads per second, etc.) of consumer computing device **160** when executing the new version of the widget is less than a threshold value and/or has reduced a threshold amount from the current or previous version of the widget. In another example, computing system **100** may receive metrics indicating that a GPU performance (e.g., video processing, memory usage, etc.) of consumer computing device **160** does not satisfy a configurable performance criterion. Computing system **100** may receive metrics indicating that execution of the new version of the widget on consumer computing device **160** results in the performance of a battery of consumer computing device **160** not satisfying criterion. Computing system **100** may receive metrics indicating that the launch time of applications implementing the new version of the widget does not satisfy criterion. Computing system **100** may receive metrics indicating consumer computing device **160** loads the new version of the widget at a rate that does not satisfy criterion. Computing system **100** may receive metrics indicating a number of error or warnings issued by either software repository **170** or consumer computing device **160** when fetching, calling, accessing, executing, etc. the new version of the widget.

(40) In response to determining that some obtained metrics do not satisfy performance criteria, computing system **100** may issue an alert or warning to developer computing device **150** indicating which performance metric is not satisfying pre-established criteria. In some instances, computing system **100** may categorize performance metrics to take a corresponding action based on the classification of a particular metric. For example, computing system **100** may categorize an obtained metric of an application load time of 5 seconds as an error that may result in an automatic roll-back of the new version of the widget. In another example, computing system **100** may identify an obtained metric of an application load time of 2 seconds as a warning and send an alert to developer computing device **150** without automatically rolling back deployment of the new version of the widget. In some examples, computing system **100** may implement any form of machine learning to intelligently determine whether obtained metrics should result in an automatic roll-back of the new version of the widget. Computing system **100** may train a machine learning model to automatically roll-back deployment of a new version of a widget according to obtained metrics and corresponding actions associated with deployment of many widgets managed by computing system **100**.

(41) FIG. 2 is a block diagram illustrating example computing environment **20** for managing software applications, in accordance with the techniques of this disclosure. In the example of FIG. 2, computing environment **20** may include computing system **200**, client devices **260A-260N**, and software repository **270**. Computing system **200**, client devices **260A-260N**, and software

repository **270** of FIG. **2** may be described as an example or alternate implementation of computing system **100**, consumer computing device **160**, and software repository **170** of FIG. **1**, respectively. One or more aspects of FIG. **2** may be described herein within the context of FIG. **1** for example purposes only. Computing system **200**, client devices **260A-260N**, and software repository **270** may be communicably coupled via network **101** of FIG. **1**, for example.

(42) Management controller **202** may include deployment module **210**, configuration file generator **208**, and Application Programming Interface (API) Gateway **220**. A developer computing device (e.g., developer computing device **150** of FIG. **1**) may upload a new version of a widget to software repository **270**. Developer computing device **150** may, for example, upload a new version of a widget as a build artifact. Developer computing device **150** may upload the new version of the widget as a build artifact by wrapping the new version of the widget in a standardized format. Developer computing device **150** uploading the new version of the widget as a build artifact ensures that the new version of the widget is compatible with all other widgets managed by computing system **200**.

(43) Configuration file generator **208** may detect whether the new version of a widget has been uploaded to software repository **270**. In some instances, configuration file generator **208** may poll software repository **270** periodically (e.g., every three seconds) or based on an event (e.g., product launch). In some examples, configuration file generator **208** may automatically poll and retrieve configuration information associated with an uploaded version of a widget. In this way, software repository **270** does not require customization to send notifications to computing system **200** that a new version of a widget has been uploaded. Configuration file generator **208** may use API Gateway **220** to retrieve configuration information associated with the new version of the widget. In some instances, configuration file generator **208** may only retrieve configuration information of widgets that have been uploaded in a standardized format (e.g., a build artifact).

(44) Configuration file generator **208** may generate a configuration file for a widget based on widget configuration information retrieved from software repository **270**. Configuration file generator **208** may generate a configuration file that is a structured representation of widget configuration information. For example, configuration file generator **208** may generate the configuration file to identify, for each widget of an application, a version of each widget and a respective location of where the corresponding version of each widget is stored. In some examples, the configuration file may include the version and one or more of a widget identifier, versions of widget dependencies, a uniform resource indicator (see FIG. **5**). Configuration file generator **208** may generate a deployment-ready configuration file for an uploaded widget responsive to the uploaded widget satisfying quality assurance and security tests. For instance, configuration file generator **208** may generate a deployment-ready configuration file for an uploaded widget only when the uploaded widget satisfies quality assurance and security tests. Configuration file generator **208** may store configuration files in storage devices **204**.

(45) In some examples, configuration file generator **208** may generate multiple configuration files for a software application. Configuration file generator **208** may generate multiple configuration files for the software application in response to a new version of a widget implemented by the software application satisfying any pre-deployment requirements (e.g., quality assurance and security tests). For example, configuration file generator **208** may maintain a configuration file for a software application implementing a current version of a widget, as well as generate a configuration file for the same software application with indications that the software application should be implemented with the new version of the widget.

(46) Management controller **202**, and more specifically deployment module **210**, may receive requests from client devices **260A-260N** to access a software application. Client devices **260A-260N** may request access to the software application with shell interfaces **262A-262N**, respectively. Shell interfaces **262A-262N** may send a request to access the software application to API Gateway **220**. API Gateway **220** may receive the request to access the software application and relay the

request to deployment module **210**.

(47) Deployment module **210** may generate a response to a request from client devices **260A-260N** to access the software application. Deployment module **210** may generate a response that includes an indication of a configuration file including widget configuration information associated with the requested software application. Deployment module **210** may generate the response that includes either a configuration file including configuration information associated with a current version of the widget or a configuration file including configuration information associated with a new version of the widget. Deployment module **210** may generate the response to specify multiple widgets used to deploy the requested software application. For example, deployment module **210** may generate a response that includes configuration files associated with a plugin and one or more utility widgets used to deploy the requested software application.

(48) A plugin may be a type of widget that includes data for a graphical user interface that outputs data associated with the one or more utility widgets underlying a software application. In some examples, a plugin may be a type of widget corresponding to a reference (e.g., uniform resource locator or URL) for a software application. A utility widget may be a type of widget that support functionality of a software application without including a frontend user interface. Utility widgets may be shared by different software applications due to the ensured compatibility according to the techniques described herein. Plugins and utility widgets may be assigned to a domain (e.g., a web domain such as domain.net) that is managed by software repository **270**. References to a software application, or widgets of the software application, may be uniform resource indicators (URIs) that append the domain managed by software repository **270**. For example, a software application may be assigned a domain of domain.net. The software application may include a widget that includes a reference to append the domain assigned to the software application, such as a reference of “/widget” that appends the domain to be “domain.net/widget.”

(49) Deployment module **210** may progressively generate a response to a request for a software application to include a new version of a widget used in the software application. Deployment module **210** may select a number of requests to receive a configuration file with configuration information associated with a new version of a widget uploaded by developer computing device **150**. For example, deployment module **210** may select a percentage of requests from client devices **260A-260N** as recipients of configuration information associated with a new version of a widget for the requested software application. Deployment module **210** may select a percentage of requests for the software application based on a window of time specified by developer computing device **150** when the new version of the widget was uploaded to software repository **270**.

(50) In some instances, after configuration file generator **208** generates a configuration file associated with a new version of a widget for a software application, deployment module **210** may only send the configuration file associated with the new version of the widget to a certain number of client devices **260A-260N**. For example, client devices **262A**, **262B**, **262C**, and **262N** may send computing system **200** a request to access a software application that uses the widget in which developer computing device **150** uploaded a new version of. Computing system **200** may receive the requests with API Gateway **220**. API Gateway **220** may relay the requests to deployment module **210**. Deployment module **210** may generate a response to each request for the software application that implements the widget with a new version.

(51) Deployment module **210** may generate responses the requests based on a time window developer computing device **150** specified when uploading the new version of the widget to software repository **270**. Deployment module **210** may divide the time window into discrete phases. For example, deployment module **210** may divide the time window into four discrete phases. Deployment module **210** may generate a response to the requests for the software application that includes configuration information associated with a current version of the widget before the time window specified by developer computing device **150** has started. During the first phase, when the time window has started, deployment module **210** may only send a response that

includes configuration information associated with the new version of the widget to client device **260A**. Deployment module **210** may monitor runtime behavior of client device **260A** executing the new version of the widget. Deployment module **210** may obtain one or more metrics associated with the runtime behavior of the new version of the widget. Deployment module **210** may determine whether the obtained metric(s) satisfy one or more configurable thresholds. In response to deployment module **210** determining all the configurable thresholds are satisfied, deployment module **210** may generate responses to send configuration information associated with the new version of the widget to client device **260A** and client device **260B** during the second phase of the time window. Deployment module **210** may generate responses that include configuration information associated with the new version of the widget for client devices **260A-260C** when the third phase of the time window has started. In some examples, deployment module **210** may generate responses associated with the new version of the widget if deployment module **210** continues to determine that the runtime behavior of the new version of the widget is satisfying configurable thresholds while executing on client devices **260A** and **260B**. During the final phase of the time window or when the time window ends, deployment module **210** may generate responses with configuration information associated with the new version of the widget for all client devices **260A-260N**.

(52) Deployment module **210** may use a machine learning model to determine a rate at which deployment module **210** should select client devices **260A-260N** as recipients of configuration information associated with a new version of a widget. For example, deployment module **210** may apply a machine learning algorithm to learn runtime behavior or characteristics associated with execution of the new version of the widget. Deployment module **210** may apply a machine learning model to output runtime characteristics associated with execution of the new version of the widget based on metrics obtained from client devices **260A-260N** executing the new version of the widget. Deployment module **210** may use any well-known machine learning techniques to intelligently ramp up or roll back deployment of a new version of a widget used for requested software applications based on the outputted runtime characteristics. In some instances, computing system **200** may train a machine learning model to automatically determine a rate of deployment of a new version of a widget or whether a new version of a widget should be rolled back. Computing system **200** may train the machine learning model with instances of previous deployment behavior of other widgets managed by computing system **200**.

(53) Shell interfaces **262A-262N** may execute a software application that includes the widget based on instructions and configuration information deployment module **210** included in a response to a request for the software application. For example, shell interface **262A** may receive configuration information and instructions to execute a requested software application with a new version of a widget. Shell interface **262A** may receive instructions from deployment module **210** to generate contextual information based on the configuration information included in the response. Shell interface **262A** may generate contextual information based on properties specified in the configuration information (e.g., host properties, child properties, etc.). Shell interface **262A** may use the generated contextual information to request software code files associated with the software application from software repository **270**. Software repository **270** may fetch the requested software code files (e.g., widget(s)) and send the files to shell interface **262A**. Shell interface **262A** may receive the software code files and execute the software application based on the contextual information generated based on the configuration information and instructions sent from deployment module **210**.

(54) The techniques may provide one or more technical advantages that realize at least one practical application. Computing system **200** may provide a common knowledge base that facilitates the development, deployment, and execution of software applications throughout the CI/CD pipeline. For example, computing system **200** may allow developer computing device **150** to proactively manage deployment of a new version of a widget used in one or more software



applications. Computing system **200** may allow a developer to configure an active deployment time window to regulate specific dates and times computing system **200** should transition consumer traffic to a new version of a widget. Computing system **200** may allow developer computing device **150** to configure operational threshold requirements. Developer computing device **150** may establish threshold operational requirements such as runtime, threads per second, etc. Developer computing device **150** may send the threshold requirements to computing system **200**. Computing system **200** may obtain metrics related to the execution of a new version of a widget. Computing system **200** may automatically roll-back, slow down, or issue an alert to developer computing device **150** when a threshold requirement is not satisfied. In this way, computing system **200** provides developer computing device **150** with the flexibility to control the speed and quality assurance of the deployment of a new version of a widget used in a software application.

(55) FIG. **3** is a block diagram illustrating example computing system **300** for managing software applications, in accordance with the techniques of this disclosure. Computing system **300** of FIG. **3** may be described as an example or alternate implementation of computing system **100** of FIG. **1** and/or computing system **200** of FIG. **2**. One or more aspects of FIG. **3** may be described herein within the context of FIGS. **1** and **2** for example purposes only.

(56) Computing system **300** may be implemented as any suitable computing system, such as one or more server computers, workstations, mainframes, appliances, cloud computing systems, and/or other computing systems that may be capable of performing operations and/or functions described in accordance with one or more aspects of the present disclosure. In some examples, computing system **300** represents a cloud computing system, server farm, and/or server cluster (or portion thereof) that provides services to other devices or systems. Computing system **300** may represent or be implemented through one or more virtualized computer instances (e.g., virtual machines, containers) of a cloud computing system, server farm, data center, and/or server cluster.

(57) In the example of FIG. **3**, computing system **300** may include one or more processors **313**, one or more communication units **315**, storage devices **304**, and management controller **302**. One or more of the devices, modules, storage areas, or other components of computing system **300** may be interconnected to allow for inter-component communications (e.g., physically, communicatively, and/or operatively). In some examples, such connectivity may be provided by communication channels (e.g., communication channels **312**), which may represent one or more of a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data.

(58) One or more communication units **315** of computing system **300** may communicate with devices external to computing system **300** by transmitting and/or receiving data, and may operate, in some respects, as both an input device and an output device. In some examples, communication units **315** may communicate with other devices over a network. In other examples, communication units **315** may send and/or receive radio signals on a radio network such as a cellular radio network. In other examples, communication units **315** of computing system **300** may transmit and/or receive satellite signals on a satellite network. Examples of communication units **215** include, but are not limited to, a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GPS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **215** may include devices capable of communicating over Bluetooth®, GPS, NFC, ZigBee®, and cellular networks (e.g., 3G, 4G, 5G), and Wi-Fi® radios found in mobile devices as well as Universal Serial Bus (USB) controllers and the like. Such communications may adhere to, implement, or abide by appropriate protocols, including Transmission Control Protocol/Internet Protocol (TCP/IP), Ethernet, Bluetooth®, NFC, or other technologies or protocols.

(59) One or more processors **313** of computing system **300** may implement functionality and/or execute instructions associated with computing system **300** or associated with one or more modules illustrated herein and/or described below. One or more processors **313** may be, may be part of,

and/or may include processing circuitry that performs operations in accordance with one or more aspects of the present disclosure. Examples of processors **313** include microprocessors, application processors, display controllers, auxiliary processors, one or more sensor hubs, and any other hardware configured to function as a processor, a processing unit, or a processing device. Computing system **300** may use one or more processors **313** to perform operations in accordance with one or more aspects of the present disclosure using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at computing system **300**.

(60) One or more processors **313** (also referred to herein as “processors **313**”) may provide an operating environment or platform for various modules described herein, which may be implemented as software, but may in some examples include any combination of hardware, firmware, and software. One or more processors **313** may execute instructions of one or more modules. Processors **313** may retrieve, store, and/or execute the instructions and/or data of one or more applications, modules, or software. Processors **313** may also be operably coupled to one or more other software and/or hardware components, including, but not limited to, one or more of the components of computing system **300** and/or one or more devices or systems illustrated as being connected to computing system **300**.

(61) Processors **313** may implement functionality and/or execute instructions within computing system **300**. For example, processors **313** may receive and execute instructions that provide the functionality of API Gateway **320**, widget monitor **306**, machine learning model **326**, contextual utility **322**, pre-deployment utilities **324**, and deployment module **310**. These instructions executed by processors **313** may cause computing system **300** to store and/or modify information within storage devices **304** or processors **313** during program execution. Processors **313** may execute instructions of API Gateway **320**, widget monitor **306**, machine learning model **326**, contextual utility **322**, pre-deployment utilities **324**, and deployment module **310** to perform one or more operations. That is API Gateway **320**, widget monitor **306**, machine learning model **326**, contextual utility **322**, pre-deployment utilities **324**, and deployment module **310** may be operable by processors **313** to perform various functions described herein.

(62) Storage devices **304** within computing system **300** may store information for processing during operation of computing system **300** (e.g., computing system **300** may store data accessed by API Gateway **320**, widget monitor **306**, machine learning model **326**, contextual utility **322**, pre-deployment utilities **324**, and deployment module **310** during execution at computing system **300**). In some examples, storage devices **304** may include temporary memory, meaning that a primary purpose of storage devices **304** are not long-term storage. Storage devices **304** of computing system **300** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if powered off. Examples of volatile memories include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories known in the art.

(63) Storage devices **304** may include one or more computer-readable storage media. Storage devices **304** may be configured to store larger amounts of information than volatile memory. Storage devices **304** may further be configured for long-term storage of information as non-volatile memory space and retain information after power on/off cycles. Examples of non-volatile memories include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Storage devices **304** may store program instructions and/or information associated with API Gateway **320**, widget monitor **306**, machine learning model **326**, contextual utility **322**, pre-deployment utilities **324**, and deployment module **310**.

(64) Storage devices **304** may represent one or more physical or virtual computer and/or storage devices that include or otherwise have access to storage media. Such storage media may include one or more of Flash drives, solid state drives (SSDs), hard disk drives (HDDs), forms of electrically programmable memories (EPROM) or electrically erasable and programmable

(EEPROM) memories, and/or other types of storage media used to support the computing system **300**. Storage devices **304** may be distributed across multiple devices or servers within a cloud computing environment provided by computing system **200**.

(65) Management controller **302** may include API Gateway **320**, widget monitor **306**, contextual utility **322**, pre-deployment utilities **324**, machine learning model **326**, and deployment module **210**. API Gateway **320** may be an API management tool that interfaces between computing devices and software repository **170** of FIG. 2. API Gateway **320** may act as an external server that forwards requests to access a software application (e.g., reverse proxy) to accept all API calls made by client devices. In this way, API Gateway **320** may allow computing system **300** to manage software applications by acting as an interface for client devices to request the appropriate components of a software application from a software repository or content delivery network.

(66) Computing system **300** may maintain configuration files that include configuration information for various versions of widgets. Computing system **3200** may intelligently regulate which version of widgets are sent to particular requests by sending appropriate configuration files associated with configuration information for particular versions of widgets.

(67) Widget monitor **206** may retrieve configuration information for a software application. Widget monitor **206** may receive configuration information specifying multiple widgets for deploying the software application. Widget monitor **206** may retrieve the configuration information via API Gateway **220**.

(68) In the example of FIG. 3, widget monitor **306** may include configuration file generator **308**. In some instances, configuration file generator **308** may generate one or more configuration files based on configuration information retrieved from widget monitor **306**. Widget monitor **306** may send the instructions to configuration file generator **308** to generate configuration files responsive to widget monitor **306** detecting a new version of a widget used to execute the software application has been uploaded. Configuration file generator **308** may generate a configuration file that includes the retrieved configuration information.

(69) In some examples, widget monitor **306** may send instructions to configuration file generator **308** to generate a pre-deployment configuration file. Configuration file generator **308** may generate a pre-deployment configuration file based on a newly uploaded widget that has not been tested for quality assurance and security compliance. Pre-deployment utilities **324** may manage pre-deployment configuration files. Pre-deployment utilities **324** may also provide software utilities (SDKs, APIs, etc.) to software administrators to run quality assurance and security tests on a newly uploaded widget. For example, pre-deployment utilities **324** may provide tools for software administrators to assign a health score to a newly uploaded widget.

(70) API Gateway **320** of management controller **302** may receive a request to execute a software application. In some instances, API Gateway **320** may obtain a request to execute a software application for managing an electronic document. Deployment module **310** may prepare a response to the request to execute the software application. Deployment module **310** may compile necessary configuration files for widgets used to execute the requested software application. Deployment module **310** may obtain the necessary configuration files from the configuration files stored in storage devices **304**.

(71) In some instances, deployment module **310** may include instructions generated by contextual utility **322** in a response to a request for a software application. Contextual utility **322** may generate instructions for loading widgets and data used to execute a software application. Contextual utility **322** may generate instructions to load a new version of a widget used to execute a software application based on configuration information associated with a version of the widget deployment module **310** selects.

(72) In some examples, deployment module **310** may apply machine learning model **326** to determine whether to output instructions to execute a current version of the widget or the next version of the widget. Deployment module **310** may input metrics obtained from a client device

executing the new version of the widget to machine learning model **326**. Machine learning model **326** may output runtime characteristics associated with the execution of the new version of the widget. Deployment module **310** may determine whether to ramp-up or roll-back deployment of the new version of the widget based on the runtime characteristics output by machine learning model **326**. Machine learning model **326** may be trained with instances of deployment behavior of multiple widgets managed by computing system **300**.

(73) FIG. **4** is a conceptual diagram illustrating an example deployment lifecycle **400** of a new version of a widget for a software application, in accordance with techniques of this disclosure. Deployment lifecycle **400** may include multiple, discrete environments associated with different stages in the deployment of a new version of a widget. In the example of FIG. **4**, deployment lifecycle **400** may include build environment **402**, post-build environment **404**, integration environment **406**, stage environment **408**, demo environment **410**, and production environment **412**. One or more aspects of FIG. **4** may be described herein within the context of FIGS. **1-3** for example purposes only.

(74) Build environment **402** may include a user of developer computing device **150** creating executable software code files for a new version of a widget used in implementing software applications. Build environment **402** may include the manual generation of executable programs by software developers using command line interface **152**, for example. Post-build environment **404** may include developer computing device **150** executing software code files of a new version of a widget created in build environment **402**. Post-build environment **404** may include a software developer locally testing a new version of a widget using command line interface **152**. Integration environment **406** may include developer computing device **150** uploading a new version of a widget to software repository **170**. In some instances, integration environment **406** may include computing system **100** determining whether the new version of the widget uploaded to software repository **170** was wrapped via a standardized format. In some examples, integration environment **406** may include quality assurance and/or security testing.

(75) Stage environment **408** may include internal testing of the new version of the widget. Stage environment **408** may include software development teams mimicking behavior of consumer computing devices **260A-260N** on the software developers' computing devices. In some instances, stage environment **408** may include performance testing. Stage environment **408** may include computing system **100** obtaining metrics from software developers executing the new version of the widget in various software application with consumer computing devices **260A-260N**. Computing system **100** may obtain metrics such as

(76) Demo environment **410** may include external testing and/or demonstration of a new version of a widget. Demo environment **410** may include software development teams deploying a new version of a software application to consumer computing devices **260A-260N**, where consumer computing devices **260** are operated by clients of the software development team. Demo environment **410** may include only deploying a new version of widget to consumer computing devices **260** associated with a select group of clients with pre-determined access to the new version of the widget.

(77) Production environment **412** may include the deployment of a new version of a widget to the general public and/or to subscribers with access to the widget. Production environment **412** may be the final stage of the deployment of the new version of the widget. Production environment **412** may include receiving requests for a software application implementing the widget, sending configuration information and instructions to execute the requested software application, and accessing the software application by executing widgets implemented in the software application based on received configuration information and instructions. In some instances, production environment **412** may include strict deployment configuration thresholds to determine whether the new version of the widget is being executed according to quality assurance and/or security standards associated with a consumer-ready software application.

(78) In accordance with aspects of this disclosure, computing system **100** may manage deployment of a new version of a widget in either demo environment **410** or production environment **412**. For example, demo environment **410** may include client devices **260** that intend to demonstrate a new version of a widget. Demo environment **410** may include client devices **260** that demonstrate particular performance metrics of a new version of a widget. Demo environment **410** may include client devices **260** that demonstrate performance of the new version of the widget to close partners, for example, who are interested in observing performance of the new version of the widget prior to releasing the new version of the widget in production environment **412**. Production environment **412** may include client devices **260** associated with consumers or customers of software applications managed by computing system **100**.

(79) In some instances, computing system **100** may roll back deployment of a new version of a widget when a new version of the widget does not satisfy criteria associated with demo environment **410** and/or production environment **412**. For example, computing system **100** may roll back deployment of a new version of a widget in demo environment **410** in response to client device **260A** executing the new version of the widget in a way that does not satisfy one or more threshold requirements. Computing system **100** may instruct all client devices **260** to execute a current or previous version of the widget. Computing system **100** may additionally move the new version of the widget back to any previous environment (e.g., stage environment **408**, integration environment **406**, etc.).

(80) FIG. 5 is a conceptual diagram illustrating an example configuration file **530**, in accordance with the techniques of this disclosure. In some instances, configuration file generator **308** of FIG. 3 may generate configuration file **530**. Configuration file generator **308** may generate configuration file **530** in response to widget monitor **306** detecting a software application, or a new version thereof, associated with configuration file **530** has been uploaded.

(81) In the example of FIG. 5, configuration file **530** may include configuration information for a software application uploaded to software repository **170** of FIG. 1, for example. Configuration file **530** may include plugin configuration **540** and widget configuration **550**. Plugin configuration **540** may include configuration information corresponding to a plugin associated with a software application. A plugin may be a type of widget that includes data for a graphical user interface that outputs data associated with one or more utility widgets underlying a software application. In some instances, a plugin may be a type of widget corresponding to a reference (e.g., uniform resource locator or URL) for a software application. Widget configuration **550** may include configuration information corresponding to a utility underlying functionality of the software application associated with configuration file **530**. A utility widget may be a type of widget that support functionality of a software application without including a frontend user interface. Utility widgets may be shared by different software applications due to the ensured compatibility according to the techniques described herein. Plugins and utility widgets may be assigned to a domain (e.g., a web domain such as domain.net) that is managed by software repository **170**. References to a software application, or widgets of the software application, may be uniform resource indicators (URIs) that append the domain managed by software repository **170**. For example, a software application may be assigned a domain of domain.net. The software application may include a widget that includes a reference to append the domain assigned to the software application, such as a reference of “/widget” that appends the domain to be “domain.net/widget.”

(82) Plugin configuration **540** may include configuration information of a plugin for a software application associated with configuration file **530**. Plugin configuration **540** may include configuration information, such as widget route **542**, authentication **544**, and plugin identification (plugin ID) **546**. An example of plugin configuration **540** is as follows:

(83) TABLE-US-00001 { “route”: “/home”, “authenticated”: { “type”: “SSO”, “clientId”: “XYZ123” }, “pluginID”: “@home-team/app” }

(84) Plugin ID **546** may include configuration information specifying a reference to the plugin

providing the user interface for the software application associated with configuration file **530**. For example, plugin ID **546** may include a uniform resource indicator (e.g., uniform resource locator or URL) for the plugin associated with the software application. Plugin ID **546** may include the configuration information used by a requesting computing device to load the plugin for the software application associated with configuration file **530**. Plugin ID **546** may include instructions for a requesting computing device to pull the plug in associated with plugin ID **546** from a software repository (e.g., software repository **170** of FIG. **1**).

(85) Authentication **544** may include configuration information indicating whether a requesting computing device has the appropriate credentials to access the software application associated with configuration file **530**. In some instances, management controller **102** of FIG. **1**, for example, may provide instructions to a requesting computing device to input credentials to access the software application. Management controller **102** may receive the inputted credentials from the requesting computing device. Management controller **102** may verify whether the inputted credentials match credentials with access to the software application associated with configuration file **530**.

Authentication **544** may include configuration information associated with whether the inputted credentials have been verified. In some examples, authentication **544** may include a Boolean value indicating that a requesting computing device has provided the verified credentials to access the software application associated with configuration file **530**.

(86) Widget route **542** may include configuration information associated with references to utility widgets that include software code for processes that execute the software application associated with configuration file **530**. Widget route **542** may be associated with a particular domain hosting the software application (e.g., domain.net). In some examples, widget route **542** may include uniform resource indicators (URIs) associated with each utility widget used to execute the software application. Widget route **542** may include a URI appending the domain hosting the software application (e.g., /widget). Widget route **542** may include a complete URI (e.g., uniform resource locator) to a utility widget associated with widget configuration **550**, such as “domain.net/widget.”

(87) Widget configuration **550** may include configuration information, such as widget identification (widget ID) **552**, version **554**, dependencies **556**, and uniform resource indicator (URI) **558**. An example of widget configuration **550** is as follows: { “widgetID”: “@prepare PDF/app”, “version”: “2.0.6” “metadata”: { “dependsOn”: [“@PDF viewer/app” ] “URIs”: [“preload.html”, “/api/me” ] } }

(88) Widget ID **552** may be a tag or identifier for a utility widget associated with widget configuration **550**. For example, widget ID **552** may be a tag assigned to the build artifact of the utility widget that was uploaded to a software repository (e.g., software repository **170** of FIG. **1**). Version **554** may include configuration information associated with a version of the utility widget used to execute the software application associated with configuration file **530**. In some instances, configuration file generator **308** of FIG. **3** may define version **554** based on whether configuration file **530** is a pre-deployment configuration file or a deployment configuration file. In examples where configuration file **530** is a pre-deployment configuration file, configuration file generator **308** may define version **554** based on a version of the utility widget that is undergoing pre-deployment testing (e.g., quality assurance and/or security testing). In examples where configuration file **530** is a deployment configuration file, configuration file generator **308** may define version **554** as a version of the utility widget that has satisfied quality assurance and security requirements.

(89) In accordance with the techniques described herein, configuration file generator **308** may generate a new widget configuration **550** that includes configuration information associated with a new version of a widget. Deployment module **310** of FIG. **3**, for example, may select one or more requests for software applications that implement the widget with the new version as recipients of the new version of the widget. Deployment module **310** may send widget configuration **550** associated with the new version of the widget in responses to requests selected as recipients of the

new version of the widget. Deployment module **310** may send widget configuration information associated with the current version of the widget in responses to requests that were not selected as recipients of the new version of the widget. Deployment module **310** may select recipients of the new version of the widget based on a number of previous requests for software applications that implement the widget. For example, deployment module **310** may select a number of recipients based on a percentage of total requests for software applications that implement the widget.

(90) Dependencies **556** may include configuration information identifying metadata specifying which, if any, utility widgets the utility widget associated with widget configuration **550** depends on. For example, widget configuration **550** may be associated with a utility widget that depends on another utility widget. In this example, dependencies **556** may include metadata identifying a reference to dependency widgets, as well as a version of the dependency widgets that the parent widget applies to execute processes of the software application associated with configuration file **530**. In some examples, computing system **100** may generate a separate configuration file in response to a new version of a dependency widget defined in dependencies **566** being uploaded. Dependencies **556** may include a uniform resource indicator (e.g., uniform resource locator) associated with the version of the dependency widget the parent widget applies in execution of the software application. In some examples, widget configuration **550** may not include dependencies **556** if the widget associated with widget configuration **550** does not depend on any other widgets.

(91) URI **558** may include configuration information on how to access the utility widget associated with widget configuration **550** from a software repository (e.g., software repository **170** of FIG. **1**). In some instances, URI **558** may represent the URL specified in widget route **542** of plugin configuration **540**. URI **558** may also represent the URL included in a widget configuration for another utility widget that depends on the widget associated with widget configuration **550**. In some examples, URI **558** may represent a website URL (e.g., “domain.net/plugin/widget”) that may be used by a consumer computing device to load the plugin and embedded widget associated with widget configurations **550** from a content delivery network or software repository. URI **558** provides utility widget specific configuration information that allows software developers and consumers to access a particular version of a software application from a content delivery network based on a version specified in version **554**. In this way, the common knowledge base platform as described herein (e.g., computing system **100** of FIG. **1**) may regulate deployment of various versions of software by intelligently sending configuration files specifying appropriate versions of widgets based on a requesting computing device. Computing system **100** may send a software developer computing device a pre-deployment configuration file that specifies untested versions of utility widgets for a requested software application, while sending consumer computing devices a deployment configuration file that specifies tested, stable versions of utility widgets for the requested software application.

(92) FIG. **6** is a block diagram illustrating an example consumer computing device **460**, in accordance with the techniques of this disclosure. Consumer computing device **460** may be an example of consumer computing device **160** of FIG. **1** or any of consumer computing devices **260A-260N** of FIG. **2**. FIG. **6** illustrates only one particular example of consumer computing device **460**, and many other examples of consumer computing device **460** may be used in other instances and may include a subset of components included in example consumer computing device **460** or may include additional components not shown in FIG. **6**.

(93) In the example of FIG. **6**, consumer computing device **460** may include processors **464**, communication units **466**, user interface (UI) components **468**, and storage devices **472**. Communication channels **476** (“COMM channel **476**”) may interconnect each of the components **464**, **466**, **468**, and **472** for inter-component communications (physically, communicatively, and/or operatively). In some examples, communication channel **476** may include a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data.

(94) In some examples, UI components **468** may be a presence-sensitive display configured to detect input (e.g., touch and non-touch input) from a user of consumer computing device **460**. UI components **468** may output information to a user in the form of a UI, which may be associated with functionality provided by consumer computing device **460**. Such UIs may be associated with computing platforms, operating systems, applications, and/or services executing at or accessible from consumer computing device **460** (e.g., electronic message applications, chat applications, Internet browser applications, mobile or desktop operating systems, social media applications, electronic games, menus, and other types of applications).

(95) Processors **464** may implement functionality and/or execute instructions within consumer computing device **460**. For example, processors **464** may receive and execute instructions that provide the functionality of shell interface **462** and operating system (OS) **474**. These instructions executed by processors **464** may cause consumer computing device **460** to store and/or modify information within storage devices **472** or processors **464** during program execution. Processors **464** may execute instructions of shell interface **462** and OS **474** to perform one or more operations. That is shell interface **462** and OS **474** may be operable by processors **464** to perform various functions described herein.

(96) In the example of FIG. 6, communication units **466** of consumer computing device **460** may communicate with one or more external devices via one or more wired and/or wireless networks by transmitting and/or receiving network signals on the one or more networks. Examples of communication units **466** include a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GNSS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **466** may include short wave radios, cellular data radios (for terrestrial and/or satellite cellular networks), wireless network radios, as well as universal serial bus (USB) controllers.

(97) In the example of FIG. 6, storage devices **472** may include operating system (OS) **474** and shell interface **462**. Storage devices **472** within consumer computing device **460** may store information for processing during operation of consumer computing device **460** (e.g., consumer computing device **460** may store data accessed by shell interface **462** and OS **474** during execution at consumer computing device **460**). In some examples, storage devices **472** may be a temporary memory, meaning that a primary purpose of storage devices **472** is not long-term storage. Storage devices **472** on consumer computing device **460** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if powered off. Examples of volatile memories include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories known in the art.

(98) Storage devices **472** may include one or more computer-readable storage media. Storage devices **472** may be configured to store larger amounts of information than volatile memory. Storage devices **472** may further be configured for long-term storage of information as non-volatile memory space and retain information after power on/off cycles. Examples of non-volatile memories include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Storage devices **472** may store program instructions and/or information associated with shell interface **462** and OS **474**.

(99) Storage devices **472** may include OS **474**. OS **474** may control the operation of components of consumer computing device **460**. For example, OS **474** may facilitate the communication of shell interface **462** with processor **464**, storage devices **472**, and communication units **466**. In some examples, OS **474** may manage interactions between software applications and a user of consumer computing device **460**. OS **474** may have a kernel that facilitates interactions with underlying hardware of consumer computing device **460** and provides a fully formed application space capable of executing a wide variety of software applications having secure partitions in which each of the



software applications executes to perform various operations. In some examples, UI components **468** may be considered a component of OS **474**.

(100) Storage devices **472** may include shell interface **462**. Shell interface **462** may include a website browser interface. In some instances, shell interface **462** may be a website browser application. Shell interface **462** may include instructions on how to request software applications from computing system **100**. In some examples, shell interface **462** may include instructions on how to execute a software application requested from computing system **100**. In the example of FIG. **6**, shell interface **462** may include contextual utility **422**. Contextual utility **422** may include instructions from computing system **100** on how shell interface **462** should execute a requested software application.

(101) In accordance with the techniques described herein, consumer computing device **460** may execute a software application for managing electronic documents. For example, shell interface **462** of computing device **460** may send a request to computing system **100** to access a software application implemented with multiple widgets. Shell interface **462** may output a text field to allow a user operating consumer computing device **460** to input a request for a software application (e.g., a user inputting a website address). Shell interface **462** may send a request to computing system **100** for access to a software application with communication units **466**. Communication units **466** may receive a response generated by computing system **100** that includes configuration information associated with widgets implemented in the software application, as well as instructions to execute the software application. Communication units **466** may relay the response generated by computing system **100** to shell interface **462** via communication channels **476**. Contextual utility **422** of shell interface **462** may process the instructions to execute the software application by generating contextual properties based on configuration information associated with the widgets implemented in the requested software application. Shell interface **462** may execute the software application by requesting widgets and data implemented in the software application from software repository **170** based on the contextual properties generated by contextual utility **422**.

(102) FIG. **7** is a conceptual diagram illustrating example contextual information **620** used to load a software application, in accordance with the techniques of this disclosure. In some instances, contextual information **620** may be generated by command line interface **152** of developer computing device **150** and/or shell interface **162** of consumer computing device **160**, of FIG. **1**. Shell interface **162**, for example, may generate contextual information **620** responsive to instructions from computing system **100** of FIG. **1**. Shell interface **162** may generate contextual information **620** based on received configuration information associated with a requested software application. For example, shell interface **162** may generate contextual information **620** based on a received configuration file (e.g., configuration file **530** of FIG. **5**) associated with a requested software application. Shell interface **162** may generate contextual information **620** based on plugin configuration **540**, for example. Shell interface **162** may generate contextual information **620** based on a utility provided by the common knowledge base platform (e.g., contextual utility **322** of FIG. **3**).

(103) In the example of FIG. **7**, contextual information **620** may include instructions to load widgets and data for a software application. Contextual information **620** may organize instructions to load widgets and data for a software application into domain properties. Contextual information **620** may include domain properties corresponding to a dependency hierarchy of a software application. For example, contextual information **620** may include platform properties **622**, host properties **624**, and children properties **626**.

(104) Platform properties **622** may include instructions to load a software application based on platform data of the requesting computing device. For example, platform properties **622** may include instructions to load the software application based on platform data associated with a deployment environment (e.g., demo environment **410** or production environment **412** of FIG. **4**) the common knowledge base platform (e.g., computing system **100** of FIG. **1**) will send

configuration information to, a browser with a command line interface (e.g., shell interface **162** of FIG. **1**) to access and execute the software application, data of a user account associated with the requesting computing device, an authentication state of the requesting computing device, a language preference of a user operating the requesting computing device, font information for presenting an electronic document, a runtime widget hierarchy, and utilities provided by the common knowledge base platform. In some examples, platform properties **622** may be generated based on plugin configuration information **540** of FIG. **5**. In addition, shell interface **162** may alter the platform data included in platform properties **622**. For example, shell interface **162** may specify font information (e.g., size of font, color of font, etc.) for presenting an electronic document.

(105) Host properties **624** may include instructions to load the software application based on a host widget. For example, host properties **624** may include instructions to load the software application based on contextual information associated with feature flags included in the host widget, a configuration API of the host widget, data used to execute the host widget, and dependent functions the host widget uses. For example, host properties **624** may include instructions based on host data that specifies a utility widget that includes processes for the software application. Host properties **624** may include instructions for shell interface **162** to load the utility widget, even before a user operating consumer computing device **160** executes the processes of the widget.

(106) Host properties **624** may be generated based on widget configuration **550** of FIG. **5**. For example, host properties **624** may be generated to include instructions based on host data associated with dependent functions the host widget uses. Host properties **624** may include instructions generated based on dependencies **556** of widget configuration **550**, indicating where to access versions of utility widget a host widget depends on. In some instances, host properties **624** may include instructions to execute a new version of a widget in response to computing system **100** selecting shell interface **162** as a recipient of the new version of the widget. Host properties **624** may include instructions to execute the current or a previous version of a widget in response to computing system **100** not selecting shell interface **162** as a recipient of the new version of the widget during the phased deployment of the new version of the widget.

(107) Host properties **624** may be generated based on host data such as metadata specifying a host widget (e.g., plugin). In some instances, a host widget may be a plugin that exposes an API for other widgets to apply in processes for the software application. Host properties **624** may allow the other widgets to use the API exposed by the host widget by including configuration, bootstrapping, and initialization information of the API.

(108) Children properties **626** may include instructions to load the software application based on children widgets used by a host widget. For example, children properties **626** may include instructions to load the software application based on children data associated with a host widget using a child widget as a wrapper or a high order component (HOC). Children properties **626** may be generated based on widget configuration **550** of FIG. **5**. For example, children properties **626** may be generated to include instructions based on contextual information associated with dependent functions the host widget uses. Children properties **626** may include instructions generated based on dependencies **556** of widget configuration **550**, indicating where to access versions of utility widget a host widget depends on. In some instances, children properties **626** may include instructions to execute a new version of a widget in response to computing system **100** selecting shell interface **162** as a recipient of the new version of the widget. Children properties **626** may include instructions to execute the current or a previous version of a widget in response to computing system **100** not selecting shell interface **162** as a recipient of the new version of the widget during the phased deployment of the new version of the widget.

(109) Contextual information **620** may allow a requesting computing device to load all widgets and data associated with a software application in parallel. In typical deployment schemes, a requesting computing device loads widgets and data sequentially by making requests to access software and data in a sequential order. Contextual information **620**, rather provides properties of the software

application based on various domains with instructions for the requesting computing device to load all widgets and data specified in contextual information **620** at one time. Contextual information **620** includes instructions to quickly load all components of a software application to provide a user of the requesting computing device a seamless experience executing various processes included in the software application.

(110) In accordance with the techniques of this disclosure, contextual information **620** may include instructions to execute either a current version of a widget or a new version of the widget.

Contextual information **620** may include instructions to execute the new version of the widget in response to computing system **100** selecting shell interface **162** at a recipient of the new version of the widget. In some instances, contextual information **620** may include instructions to execute the current version of the widget in response to computing system **100** determining that the deployment of the new version of the widget should be rolled back based on performance metrics associated with the execution of the new version of the widget not satisfying criteria. For example, contextual information **620** may initially include instructions to execute the new version of the widget. Shell interface **162** may execute a software application including the new version of the widget based on contextual information **620**. Shell interface **162** may send computing system **100**, via API Gateway **220**, performance metrics associated with the execution of the software application and/or the new version of the widget. In response to computing system **100** determining the obtained metrics do not satisfy performance, quality assurance, security, or other types of requirements, computing system **100** may send instructions to shell interface **162** to update contextual information **620**. Computing system **100** may send instructions to shell interface **162** to update contextual information **620** with instructions to execute the current or a previous version of the widget that computing system **100**.

(111) In some instances, contextual information **620** may include instructions to execute the current version of the widget in response to computing system **100** not selecting shell interface as a recipient of the new version of the widget in a particular phase. Computing system **100** may subsequently select shell interface **162** as a recipient of instructions to execute the new version of the widget in a later phase. Computing system **100** may send instructions to shell interface **162** to update contextual information **620** with instructions to execute the new version of the widget.

(112) FIG. **8** is a block diagram illustrating an example developer computing device **450**, in accordance with the techniques of this disclosure. Developer computing device **450** may be an example of developer computing device **150** of FIG. **1**. FIG. **8** illustrates only one particular example of developer computing device **450**, and many other examples of developer computing device **450** may be used in other instances and may include a subset of components included in example developer computing device **450** or may include additional components not shown in FIG. **8**.

(113) In the example of FIG. **8**, developer computing device **450** may include processors **454**, communication units **456**, user interface (UI) components **458**, and storage devices **442**. Communication channels **446** (“COMM channel **446**”) may interconnect each of the components **442**, **454**, **456**, and **458** for inter-component communications (physically, communicatively, and/or operatively). In some examples, communication channel **446** may include a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data.

(114) In some examples, UI components **458** may be a presence-sensitive display configured to detect input (e.g., touch and non-touch input) from a user of developer computing device **450**. UI components **458** may output information to a user in the form of a UI, which may be associated with functionality provided by developer computing device **450**. Such UIs may be associated with computing platforms, operating systems, applications, and/or services executing at or accessible from developer computing device **450** (e.g., electronic message applications, chat applications, Internet browser applications, mobile or desktop operating systems, social media applications,

electronic games, menus, and other types of applications).

(115) Processors **454** may implement functionality and/or execute instructions within developer computing device **450**. For example, processors **454** may receive and execute instructions that provide the functionality of command line interface **452** and operating system (OS) **444**. These instructions executed by processors **454** may cause developer computing device **450** to store and/or modify information within storage devices **442** or processors **454** during program execution. Processors **454** may execute instructions of command line interface **452** and OS **444** to perform one or more operations. That is command interface **452** and OS **444** may be operable by processors **454** to perform various functions described herein.

(116) In the example of FIG. **8**, communication units **456** of developer computing device **450** may communicate with one or more external devices via one or more wired and/or wireless networks by transmitting and/or receiving network signals on the one or more networks. Examples of communication units **456** include a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GNSS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **456** may include short wave radios, cellular data radios (for terrestrial and/or satellite cellular networks), wireless network radios, as well as universal serial bus (USB) controllers.

(117) In the example of FIG. **6**, storage devices **442** may include operating system (OS) **444** and command line interface **452**. Storage devices **442** within developer computing device **450** may store information for processing during operation of developer computing device **450** (e.g., developer computing device **450** may store data accessed by command line interface **452** and OS **444** during execution at developer computing device **450**). In some examples, storage devices **442** may be a temporary memory, meaning that a primary purpose of storage devices **442** is not long-term storage. Storage devices **442** on developer computing device **450** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if powered off. Examples of volatile memories include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories known in the art.

(118) Storage devices **442** may include one or more computer-readable storage media. Storage devices **442** may be configured to store larger amounts of information than volatile memory. Storage devices **442** may further be configured for long-term storage of information as non-volatile memory space and retain information after power on/off cycles. Examples of non-volatile memories include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Storage devices **442** may store program instructions and/or information associated with command line interface **452** and OS **444**.

(119) Storage devices **442** may include OS **444**. OS **444** may control the operation of components of developer computing device **450**. For example, OS **444** may facilitate the communication of command line interface **452** with processors **454**, storage devices **442**, and communication units **456**. In some examples, OS **444** may manage interactions between software applications and a user of developer computing device **450**. OS **444** may have a kernel that facilitates interactions with underlying hardware of developer computing device **450** and provides a fully formed application space capable of executing a wide variety of software applications having secure partitions in which each of the software applications executes to perform various operations. In some examples, UI components **458** may be considered a component of OS **444**.

(120) Storage devices **442** may include command line interface **452**. Command line interface **452** may include a shell interface, a software development kit, an application programming interface, or any other interface a software developer can use to create and execute software code. In the example of FIG. **8**, command line interface **452** may include pre-deployment utilities **424** and contextual utility **422**. Contextual utility **422** may include instructions received from computing

system **100** on how to execute software applications implemented by multiple widgets. Contextual utility **422** may include similar functionality to contextual utility **422** of FIG. **6**.

(121) In accordance with the techniques described herein, command line interface **452** may upload a new version of a widget to software repository **170**. Pre-deployment utilities **424** may include instructions received from computing system **100**. Pre-deployment utilities **424** of command line interface **452** may output an option to a user, via UI components **458**, to input deployment configuration associated with thresholds representing required performance metrics associated with a particular environment (e.g., demo environment **410** or production environment **412** of FIG. **4**). Examples of deployment configurations are discussed throughout this disclosure, such as in the description of FIG. **9** below. Pre-deployment utilities **424** may include instructions for quality assurance and/or security testing command line interface **452** may perform on new developed widgets.

(122) FIG. **9** is a conceptual diagram illustrating an example deployment configuration **650** for a new version of a widget, in accordance with techniques of this disclosure. FIG. **9** is discussed with FIGS. **1-8** for example purposes only. In the example of FIG. **9**, deployment configuration **650** may include environment **652**, widget version information **654**, metric thresholds **656**, and time window **658**.

(123) Environment **652** may include an environment (e.g., demo environment **410** or production environment **412** of FIG. **4**) a developer wants to deploy a new version of a widget to. For example, pre-deployment utilities **424** of FIG. **8** may include instructions to output, via UI components **458**, an option for a user of developer computing device **450** to select either a demo environment or production environment. In the example of FIG. **9**, deployment configuration **650** may include an indication of a software developer selecting a demo environment.

(124) Widget version information **654** may include a reference to a widget associated with the new version of the widget developer computing device **450** will deploy. Widget version information **654** may include a widget route specifying a URI associated with the widget and a version number associated with the new version of the widget that will be deployed. In some instances, pre-deployment utilities of FIG. **8** may include instructions to output, via UI components **458**, an option for a user of developer computing device **450** to input values for 'widget route' and 'version.' In the example of FIG. **9**, deployment configuration **650** may include an indication for deploying a new version '1.2.3' of a 'PDFviewer' widget that may be located at 'domain.net/pdfViewer.'

(125) Metric thresholds **656** may include configuration threshold criteria associated with expectations for the runtime execution of a new version of a widget defined in widget version information **654**. In some instances, pre-deployment utilities of FIG. **8** may include instructions to output, via UI components **458**, an option for a user of developer computing device **450** to optionally change metric threshold criteria associated with runtime execution of the new version of the widget defined in widget version information **654**. If developer computing device **450** does not receive any input from the user operating developer computing device **450** indicating a desire to change execution threshold criteria, metric thresholds **656** may include widget execution threshold criteria established by a software application administrator (e.g., an administrator managing computing system **100** of FIG. **1**). In the example of FIG. **9**, metric thresholds **656** may enable a user of developer computing device **450** to 'change performance variables,' 'change allowable number of errors or warnings,' 'change application launch time threshold,' and/or 'change widget load time threshold.' Metric thresholds **656** may allow a user to change threshold criteria associated performance variables such as computer processing unit (CPU) performance (e.g., clock cycles, power consumption, etc.), memory performance, graphics processing unit (GPU) performance, and/or battery performance. Metric thresholds **656** may allow a user to configure a threshold criterion associated with a number of allowable errors or warnings that may occur during execution of the new version of the widget. For example, metric thresholds **656** may define a minimum

number of allowable compilation errors or warnings that may occur during the execution of the new version of the widget on a client device (e.g., consumer computing device **160** of FIG. **1**) (126) Metric thresholds **656** may allow a user to change threshold criteria associated with application launch time. For example, metric thresholds **656** may define a maximum allowable application launch time threshold defining a criterion for the maximum amount of time an application implementing the new version of the widget should launch on a client device (e.g., consumer computing device **160** of FIG. **1**). Metric thresholds **656** may allow a user to change threshold criteria associated with widget load time. For example, metric thresholds **656** may define a maximum allowable widget load time threshold defining a criterion for the maximum amount of time the new version of the widget should take to load on a client device (e.g., consumer computing device **160** of FIG. **1**).

(127) In response to an application implementing the new version of the widget not satisfying the application launch time criteria, computing system **200**, or more specifically deployment module **210** of FIG. **2**, may pause deployment of the new version of the widget, roll-back deployment of the new version of the widget, and/or issue a warning to developer computing device **450** that uploaded the new version of the widget. In response to the execution of the new version of the widget not satisfying the widget load time criteria defined in metric thresholds **656**, deployment module **210** may pause deployment of the new version of the widget, roll-back deployment of the new version of the widget, and/or issue a warning to developer computing device **450** that uploaded the new version of the widget. Similarly, in response to execution of the new version of the widget not satisfying criteria associated with performance variables and/or the allowable number of errors or warnings, deployment module **210** may pause deployment of the new version of the widget, roll-back deployment of the new version of the widget, and/or issue a warning to developer computing device **450** that uploaded the new version of the widget. Deployment module **210** may obtain metrics associated with metric thresholds **656** from a client device that has been selected and is executing the new version of the widget in any software application that implements the new version of the widget.

(128) Time window **658** may include a specific period of time for the progressive deployment of the new version of the widget specified in deployment configuration **650**. Time window **658** may define the period of time the new version of the widget specified in widget version information **654** will be progressively deployed to client devices requesting software applications that implement the widget. In some instances, pre-deployment utilities of FIG. **8** may include instructions to output, via UI components **458**, an option for a user of developer computing device **450** to input values for a 'start time' and an 'end time' for progressive deployment of the new version of the widget. Time window **658** may define a start time and end time according to a UNIX timestamp. In the example of FIG. **9**, time window **658** defines a start time with a UNIX timestamp of '1682348458' which corresponds to Apr. 24, 2023 11:00 AM Eastern Time. Time window **658**, in the example of FIG. **9**, defines an end time with a UNIX timestamp of '1682352058' which corresponds to Apr. 24, 2023 12:00 PM Eastern Time.

(129) In accordance with the techniques described herein, deployment module **210** may apply deployment configuration **650** when generating responses to requests for software applications implementing a widget associated with the widget defined in widget version information **654**. For example, client devices **260A-260N** may be operated by users specific to demo environment **410** of FIG. **4**. Client devices **260A-260N** may request a software application that implements a widget associated with a new version of a widget uploaded by developer computing device **150**. Prior to the time defined as the start time in time window **658** (e.g., prior to Apr. 24, 2023 11:00 AM Eastern Time), deployment module **210** sends configuration information and instructions to execute the software application with a current version of the widget. After the start time defined in time window **658**, deployment module **210** may select one or more requests for software applications that implement the widget as recipients of the new version of the widget. For example, deployment

module **210** may determine and/or refresh a number of requests for software applications that implement the widget every five seconds during the time period defined in time window **658**. As time approaches the end time defined in time window **658**, deployment module **210** may progressively increase a percentage of the number of requests that may receive instructions to execute the new version of the widget.

(130) Throughout the progressive deployment of the new version of the widget, deployment module **210** may obtain metrics from client devices **260A-260N** that are currently executing the new version of the widget. Deployment module **210** may determine whether the obtained metrics satisfy criteria defined in metric thresholds **656**. In response to the obtained metrics satisfying all the criteria defined in metric thresholds **656**, deployment module **210** may continue to increase the percentage of requests selected as recipients of the new version of the widget until all requests are instructed to execute the new version of the widget at or shortly after the time defined in the end time of time window **658**. In response to the obtained metrics not satisfying at least one criterion defined in metric thresholds **656**, deployment module **210** may pause selecting requests as recipients of the new version of the widget, send responses to all requests that include a current version of the widget, and/or issue a warning to developer computing device **150** that uploaded the new version of the widget.

(131) FIG. **10** is a flow chart illustrating an example mode of operation for managing deployment of software applications, in accordance with the techniques of this disclosure. FIG. **10** is discussed with FIGS. **1-9** for example purposes only. In the example of FIG. **10**, computing system **100**, software repository **170**, developer computing device **150**, and consumer computing device **160** may correspond to computing system **100**, software repository **170**, developer computing device **150** and consumer computing device **160** of FIG. **1**.

(132) Developer computing device **150** may upload a new version of a widget for a software application to software repository **170** (**702**). Developer computing device **150** may upload a new version of the widget in demo environment **410** or production environment **412** of FIG. **4**.

Developer computing device **150** may upload a new version of a widget that was tested using pre-deployment utilities provided by computing system **100**. Developer computing device **150** may upload a new version of the widget for the software application to software repository **170** in response to the new version of the widget satisfying pre-deployment testing requirements.

Developer computing device **150** may upload the new version of the widget with a command line interface (e.g., command line interface **152**). In some instances, command line interface **152** may upload the new version of the widget using a utility provided by computing system **100**. Command line interface **152** may use the utility to upload a build artifact that converts a new version of the widget into a standardized format. Command line interface **152** may send the build artifact of the new version of the widget to software repository **170** via network **101**. Software repository **170** may store the new version of the widget for the software application (**704**). Software repository **170** may accordingly store the build artifact of the new version of the widget to help to ensure that all software applications stored on software repository **170** are compatible.

(133) Computing system **100** may generate a configuration file for the new version of the widget for the software application (**706**). In some instances, management controller **302** may retrieve the configuration information for the new version of the widget. For example, widget monitor **306** of management controller **302** may continuously poll software repository **170** for new versions of the widget. In response to widget monitor **306** detecting software repository **170** has stored the new version of the widget, widget monitor **306** may retrieve the configuration information associated with the new version of the widget. In some examples, configuration file generator **308** may generate a deployment configuration file associated with the software application that includes the new version of the widget. Configuration file generator **308** may generate a deployment configuration file such as configuration file **530** of FIG. **5**. Configuration file generator **308** may include configuration information associated with the new version of the widget as part of widget

configuration **550**. In some examples, the new version of the widget may be specified in dependencies **556** of widget configuration **550**.

(134) In the example of FIG. **10**, consumer computing device **160** may request access to a software application that implements the widget (**708**). Consumer computing device **160** may request access to the software application using shell interface **162**. Consumer computing device **160** may be a consumer computing device requesting access to the software application to manage electronic documents. Computing system **100** may receive the request to access the software application. Computing system may receive the request to access the software application with API Gateway **220**. Computing system **100**, or more specifically deployment module **210**, may select the request as a recipient of the new version of the widget based on a number of previous requests for the widget (**710**). In some instances, deployment module **210** may include a preliminary threshold to determine whether the new version of the widget is compatible with the ecosystem provided by computing system **100**. Deployment module **210** may determine whether protocols and/or libraries applied in the new version of the widget are consistent with other protocols and/or libraries applied in other widgets managed by computing system **100**. For example, computing system **100** may support React **17**, but the new version of the widget applies React **18**. Deployment module **210** may detect the discrepancy of the version of React applied and issue a warning to developer computing device **150** to alter the new version of the widget to be compatible with the ecosystem supported by computing system **100**.

(135) Computing system **100** may send a response to the request that includes the configuration file for the new version of the widget and instructions to execute the software application with the new version of the widget (**712**). Computing system **100** may generate a response to the request that includes the deployment configuration file (e.g., configuration file **530** of FIG. **5**) for the software application that includes the new version of the widget in, for example, widget configuration **550** of FIG. **5**. Computing system **100** may also generate a response to the request that includes instructions to execute the software application. Computing system **100** may send the response to consumer computing device **160** using API Gateway **220**, for example. Consumer computing device **160** may receive the response with the deployment configuration file and instructions to execute the software application with the new version of the widget (**714**). Consumer computing device **160** may receive the response with shell interface **162**. Consumer computing device **160** may receive the response that includes the configuration information associated with the software application and the new version of the widget.

(136) Consumer computing device **160**, or more particularly shell interface **162**, may request widgets for the software application based on the deployment configuration file and instructions received from computing system **100** (**716**). Consumer computing device **160** may send the request to software repository **170** via network **101**, for example. Consumer computing device **160** may send a request to fetch widgets based on reference indications (URIs or URLs) included in the deployment configuration file. Software repository **170** may fetch the widgets based on the request sent from consumer computing device **160** (**718**). Software repository **170** may fetch the widgets for the software application based on the reference indications included in the request sent by consumer computing device **160**. Software repository **170** may send consumer computing device **160** the widgets for the software application (**720**). Software repository **170** may send the widgets to consumer computing device **160** via network **101**. Consumer computing device **160** may load the received widgets (**722**). Consumer computing device **160** may load and execute the received widgets according to the instructions received from computing system **100**. In some instances, consumer computing device **160** may send computing system **100** metrics associated with loading widgets for the software application. Consumer computing device **160** may send the runtime execution metrics to computing system **100** via API Gateway **220**, for example. Computing system **100** may compare the obtained metrics to criteria defined in metric thresholds **656** of FIG. **9**. Computing system **100** may determine whether to continue deployment, roll-back deployment,



and/or issue a warning to developer computing device **150** responsive to the comparison of obtained metrics to criteria defined in metric thresholds **656**.

(137) FIG. **11** is a flow chart illustrating an example mode of operation for deploying a new version of a widget, in accordance with the techniques of this disclosure. FIG. **10** is discussed with FIGS. **1-9** for example purposes only. In the example of FIG. **11**, computing system **100** and developer computing device **150** may correspond to computing system **100** and developer computing device **150** of FIG. **1**, respectively. Consumer computing devices **160A-160C** may each correspond to consumer computing device **160** of FIG. **1** and/or client devices **160A-160N** of FIG. **2**.

(138) Once developer computing device **150** uploads a new version of a widget to software repository **170**, computing system **100** may instruct developer computing device **150** to input deployment configuration (e.g., deployment configuration **650** of FIG. **9**). Developer computing device **150** may send information associated with deployment configuration **650** to computing system **100** (**802**). At the start time defined in time window **658** of deployment configuration **650**, computing system **100** may receive requests **804A-804C** for software applications that implement the widget. Request **804A** may be sent by consumer computing device **160A**, request **804B** may be sent by consumer computing device **160B**, request **804C** may be sent by consumer computing device **160C**. Computing system **100** may determine a number of requests for the widget with the new version (**806**). In the example of FIG. **11**, computing system **100** may determine there are three requests for software applications associated with the widget. Computing system **100** may select one or more recipients for the new version of the widget based on the number of requests for the widget and the current time in relation to time window **658**. Because computing system **100** received requests **804A-804C** at the start time defined in time window **658**, computing system **100** may only select consumer computing device **160A** as a recipient for the new version of the widget.

(139) Computing system **100** may send consumer computing device **160A** instructions to execute the new version of the widget (**810A**). Computing system **100** may send consumer computing devices **160B** and **160C** instructions to execute the current version of the widget (**810B** and **810C**). Consumer computing device **160A** may send computing system **100** metrics associated with runtime execution of the new version of the widget (**812**). Consumer computing device **160A** may send metrics to computing system **100**, such as gate failures, CPU performance, memory performance, GPU performance, battery performance, application launch time, widget load time, number of errors or warnings during runtime execution of the new version of the widget, uncaught bugs, API issues, critical security issues, a particular combination of widget performing poorly, etc. Computing system **100** may determine whether metrics satisfy corresponding criteria (**814**). Computing system **100** may compare the obtained metrics to corresponding criteria defined in metric thresholds **656** and/or corresponding criteria established by a software administrator associated with computing system **100**.

(140) As time approaches the end time defined in time window **658**, computing system **100** may receive requests **816A** and **816B** for software applications that implement the widget associated with the new version. In response to the obtained metrics satisfying corresponding criteria, computing system **100** may send responses **818A** and **818B** that include instructions to execute the new version of the widget. In response to the obtained metrics not satisfying at least one criterion, computing system **100** may send responses **818A** and **818B** instructions to execute the current version of the widget. In this way, computing system **100** may avoid performance degradation associated with the new version of the widget affecting more than a few users of consumer computing devices **160**.

(141) FIG. **12** is a flow chart illustrating an example mode of operation for managing the deployment of a new version of a widget, in accordance with the techniques of this disclosure. FIG. **10** is discussed with FIGS. **1-9** for example purposes only. In the example of FIG. **12**, computing system **100** may determine, for a time window, a current version of a widget and a next version of the widget to implement a software application (**910**). In response to receiving a request for the

software application, computing system **100** may output instructions to execute the current version of the widget (**912**). In response to receiving a second request for the software application after the first request, computing system **100** may output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget (**914**). Computing system **100** may receive metrics associated with execution of the next version of the widget (**916**).

(142) For processes, apparatuses, and other examples or illustrations described herein, including in any flowcharts or flow diagrams, certain operations, acts, steps, or events included in any of the techniques described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the techniques). Moreover, in certain examples, operations, acts, steps, or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially. Further certain operations, acts, steps, or events may be performed automatically even if not specifically identified as being performed automatically. Also, certain operations, acts, steps, or events described as being performed automatically may be alternatively not performed automatically, but rather, such operations, acts, steps, or events may be, in some examples, performed in response to input or another event.

(143) The detailed description set forth below, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(144) In accordance with one or more aspects of this disclosure, the term “or” may be interpreted as “and/or” where context does not dictate otherwise. Additionally, while phrases such as “one or more” or “at least one” or the like may have been used in some instances but not others; those instances where such language was not used may be interpreted to have such a meaning implied where context does not dictate otherwise.

(145) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored, as one or more instructions or code, on and/or transmitted over a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of a computer program from one place to another (e.g., pursuant to a communication protocol). In this manner, computer-readable media generally may correspond to (1) tangible computer-readable storage media, which is non-transitory or (2) a communication medium such as a signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

(146) By way of example, and not limitation, such computer-readable storage media can include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if instructions are transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless

technologies such as infrared, radio, and microwave are included in the definition of medium. It should be understood, however, that computer-readable storage media and data storage media do not include connections, carrier waves, signals, or other transient media, but are instead directed to non-transient, tangible storage media. Disk and disc, as used, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

(147) Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the terms “processor” or “processing circuitry” as used herein may each refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described. In addition, in some examples, the functionality described may be provided within dedicated hardware and/or software modules. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(148) The techniques of this disclosure may be implemented in a wide variety of devices or apparatuses, including a wireless handset, a mobile or non-mobile computing device, a wearable or non-wearable computing device, an integrated circuit (IC) or a set of ICs (e.g., a chip set). Various components, modules, or units are described in this disclosure to emphasize functional aspects of devices configured to perform the disclosed techniques, but do not necessarily require realization by different hardware units. Rather, as described above, various units may be combined in a hardware unit or provided by a collection of interoperating hardware units, including one or more processors as described above, in conjunction with suitable software and/or firmware.

## Claims

1. A system for deploying a software application associated with document management, the system comprising processing circuitry implemented by one or more processors, the processing circuitry having access to a memory and configured to: determine, for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget; in response to receiving a first request for the software application, output instructions to execute the current version of the widget; and in response to receiving a second request for the software application after the first request, output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request.

2. The system of claim 1, wherein the first request is sent by a first computing device and the second request is sent by a second computing device, and wherein the processing circuitry is further configured to: receive, from the second computing device, metrics associated with execution of the next version of the widget.

3. The system of claim 2, wherein the metrics comprise at least one of: a gate failure, hardware performance of the second computing device executing the next version of the widget, a software application launch time, a widget load time, or one or more error logs.

4. The system of claim 2, wherein the processing circuitry is further configured to: in response to receiving a third request for the software application, output the instructions to execute the next version of the widget in response to a determination that the metrics satisfies one or more criteria, wherein the one or more criteria are associated with a state of an environment sending the first request and the second request, wherein the state of the environment comprises an operating

system.

5. The system of claim 2, wherein the processing circuitry is further configured to: determine the metrics associated with execution of the next version of the widget do not satisfy one or more criteria; and in response to receiving a third request for the software application and the determination that the metrics associated with execution of the next version of the widget do not satisfy the one or more criteria, output instructions to execute the current version of the widget.

6. The system of claim 5, wherein, in response to a determination that the metrics associated with execution of the next version of the widget do not satisfy the one or more criteria, the processing circuitry is further configured to: output a warning to a developer computing device associated with the next version of the widget.

7. The system of claim 2, wherein, the processing circuitry is further configured to: determine the metrics associated with execution of the next version of the widget satisfy one or more criteria; and in response to receiving a third request for the software application from a third computing device and the determination that the metrics associated with execution of the next version of the widget satisfy the one or more criteria, output instructions to execute the next version of the widget based on the number of responses to previous requests that select the current version of the widget and the time window.

8. The system of claim 7, wherein the processing circuitry is further configured to: in response to receiving a fourth request for the software application from a fourth computing device and a fifth request for the software application from a fifth computing device, output instructions, to the fourth computing device and the fifth computing device, to execute the next version of the widget based on the number of responses to previous requests that select the current version of the widget.

9. The system of claim 2, wherein the processing circuitry is further configured to: provide the metrics associated with execution of the next version of the widget as an input to a machine learning model, wherein the machine learning model is configured to output runtime characteristics associated with execution of the next version of the widget; and determine, based on the runtime characteristics, whether to output instructions to execute the current version of the widget or the next version of the widget in response to receiving a subsequent request for the widget.

10. The system of claim 1, wherein the first call and the second call are associated with a demonstration environment or a production environment, wherein the processing circuitry is configured to: output the instructions to execute the next version of the widget responsive to determining the next version of the widget satisfies at least one preliminary threshold for a respective environment.

11. The system of claim 1, wherein the processing circuitry is further configured to: refresh the number of previous requests within the time window; determine a percentage of requests executing the current version of the widget; and output instructions to execute the next version of the widget based on the percentage of requests executing the current version of the widget.

12. A method for deploying a software application associated with document management comprising: determining, by a computing system and for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget; in response to receiving a first request for the software application, outputting, by the computing system, instructions to execute the current version of the widget; and in response to receiving a second request for the software application after the first request, outputting, by the computing system, instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request.

13. The method of claim 12, wherein the first request is sent by a first computing device and the

second request is sent by a second computing device, and wherein the method further comprises: receiving, from the second computing device, metrics associated with execution of the next version of the widget.

14. The method of claim 13, wherein the metrics comprise at least one of: a gate failure, hardware performance of the second computing device executing the next version of the widget, a software application launch time, a widget load time, or one or more error logs.

15. The method of claim 13, further comprising: in response to receiving a third request for the software application, outputting instructions to execute the next version of the widget in response to a determination that the metrics satisfies one or more criteria, wherein the one or more criteria are associated with a state of an environment sending the first request and the second request, wherein the state of the environment comprises an operating system.

16. The method of claim 12, wherein the first call and the second call are associated with a demonstration environment or a production environment, and wherein outputting the instructions to execute the next version of the widget comprises: outputting the instructions to execute the next version of the widget responsive to determining the next version of the widget satisfies at least one preliminary threshold for a respective environment.

17. The method of claim 12, further comprising: refreshing the number of previous requests within the time window; determining a percentage of requests executing the current version of the widget; and outputting instructions to execute the next version of the widget based on the percentage of requests executing the current version of the widget.

18. Computer-readable storage media comprising instructions that, when executed, configure processing circuitry of a computing system to: determine, for a time window, a current version of a widget and a next version of the widget to implement a software application of the one or more software applications, wherein a repository is configured to provide the current version of the widget in response to a first call for the current version of the widget and provide the next version of the widget in response to a second call for the next version of the widget; in response to receiving a first request for the software application, output instructions to execute the current version of the widget; and in response to receiving a second request for the software application after the first request, output instructions to execute the next version of the widget based on a number of responses to previous requests that select the current version of the widget, the previous requests including the first request.

19. The computer-readable storage media of claim 18, wherein the first request is sent by a first computing device and the second request is sent by a second computing device, and wherein the processing circuitry is further configured to: receive, from the second computing device, metrics associated with execution of the next version of the widget.

20. The computer-readable storage media of claim 18, wherein the processing circuitry is further configured to: refresh the number of previous requests within the time window; determine a percentage of requests executing the current version of the widget; and output instructions to execute the next version of the widget based on the percentage of requests executing the current version of the widget.

---